

Nothing Crashed

Team Members: _____

Part 1 — The Report

Riverton is an invented city, and its bus system and data are invented too. A summer intern analyzed every trip Riverton Transit ran in 2025 and turned in this report:

Riverton Transit — 2025 Performance Summary (draft)

1. The Blue Line is our worst line. It had more late trips (more than 5 minutes behind schedule) than any other line.
2. Riverton Transit advertises that 80% of trips arrive within 5 minutes of schedule. In a sample of 80 trips, only 51 did. A simulation gives a p-value of 0.0, so the 80% claim is false.

1. Before you see any code: which claim do you trust least? Each team member circles one on their own, then compare.

Me: 1 2 Teammates chose: _____

2. For the claim you circled, name one specific thing that could have gone wrong.

STOP. Every team member answers 1 and 2 before anyone opens the notebook.

Part 2 — The Intern's Notebook

The notebook has the intern's code exactly as written. Every cell runs without an error. **Write each prediction before you run the cell it is about.**

Exhibit A — Claim 1

```
late_trips = trips.where('minutes late', are.above(5))
late_trips.group('line')
```

3. *Predict:* The output will give a count of late trips for each line. Give a reason one line could have the largest count without being the worst line.

4. A.3, by hand: Blue's fraction late = _____ late trips ÷ _____ total trips = _____

Then from the notebook, the fraction late for every line:

Blue _____ Red _____ Green _____ Gold _____

Claim 1: STANDS FALLS The worst line is _____

Exhibit B — Claim 2, the sample

```
sample = trips.take(np.arange(80))
observed = np.count_nonzero(sample.column('minutes late') <= 5)
```

5. **Predict:** The table is sorted by date and covers all 12 months. Roughly which dates will the intern's 80 trips come from? Why would that matter for a bus system?

6. **After B.2 and B.3:** The intern's sample ran from _____ to _____.

with_replacement = TRUE FALSE because

Your random sample: _____ of 80 trips on time.

Exhibit C — Claim 2, the test

```
simulated = make_array()
for i in np.arange(10000):
    one = claim.sample_from_distribution('Chance', 80).column('Chance
sample').item(0)
simulated = np.append(simulated, one)

p_value = np.count_nonzero(simulated <= observed) / 10000
```

7. **Trace before you run (after C.1):** Here is the intern's loop shrunk to 3 repetitions. Suppose the three simulated on-time counts come out 63, then 66, then 61. Fill in what simulated holds at each moment.

Moment	one	simulated
before the loop	—	[] (empty)
after repetition i = 0	63	
after repetition i = 1	66	
after repetition i = 2	61	
after the np.append line	61	

So with 10,000 repetitions, simulated will hold _____ number(s).

8. **After C.4:** Your team's p-value with the loop fixed:

p-value: _____ Claim 2: STANDS FALLS

Part 3 — The Verdict

9. As the intern wrote it, what is the *largest* p-value the test in Exhibit C could ever produce? What does that tell you about every conclusion that code could have reached?

Hint: simulated holds one number. How many of its values can be $\leq$ observed? Then divide by 10000.

10. Rewrite Claim 2 as one honest sentence, using your team's numbers.

If You Finish Early

- A. Every team drew a different random sample. Compare your p-value with another team's. If one team in the room got a p-value below 0.05, what should the class conclude about Claim 2?

- B. The intern's table holds every trip Riverton ran in 2025. To check Claim 2, do you need a sample — or a hypothesis test — at all? If a test still makes sense, what question is it answering?